

Monte Carlo Electron Track Structure Calculations in Liquid Water Using a New Model Dielectric Response Function

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Monte Carlo track structure codes provide valuable information for understanding radiation effects down to the DNA level, where experimental measurements are most difficult or unavailable. It is well recognized that the performance of such codes, especially at low energies and/or subcellular level, critically depends on the reliability of the interaction cross sections that are used as input in the simulation. For biological media such as liquid water, one of the most challenging issues is the role of condensed-phase effects. For inelastic scattering, such effects can be conveniently accounted for through the complex dielectric response function of the media. However, for this function to be useful it must fulfill some important sum rules and have a simple analytic form for arbitrary energy- and momentum-transfer. The Emfietzoglou-Cucinotta-Nikjoo (ECN) model offers a practical, self-consistent and fully analytic parameterization of the dielectric function of liquid water based on the best available experimental data. An important feature of the ECN model is that it includes, in a phenomenological manner, exchange and correlation effects among the screening electrons, thus, going beyond the random-phase approximation implicit in earlier models. In this work, inelastic cross sections beyond the plane wave Born approximation are calculated for low-energy electrons (10 eV–10 keV) based on the ECN model, and used for Monte Carlo track structure simulations of physical quantities relevant to the microdosimetry of low-energy electrons in liquid water. Important new developments in the physics of inelastic scattering are discussed and their effect on electron track structure is investigated by a comparison against simulations (under otherwise identical conditions) using the Born approximation and a simpler form of the dielectric function based on the Oak Ridge National Laboratory model. The results reveal that both the dielectric function and the corrections to the Born approximation may have a sizeable effect on track structure calculations at the nanometer scale (DNA level), where the details of inelastic

scattering and the role of low-energy electrons are most critical. © 2017 by Radiation Research Society

INTRODUCTION

Monte Carlo track structure (MCTS) simulations currently offer the most sophisticated theoretical tool for studying radiation effects at the cellular and DNA level where experimental measurements are most difficult or unavailable (1, 2). Contrary to the macroscopic Monte Carlo codes, which, due to limitations of the condensed history technique, are ill-suited for electron transport well below 1 keV, MCTS codes are capable of simulating collision-by-collision full-slowing-down electron tracks in matter (3–5). Thus, MCTS codes enable the simulation of sub-keV secondary electrons, including the so-called track-ends, which are associated with high radiobiological effectiveness (6, 7). In recent years, the degree of sophistication of MCTS codes has expanded beyond the simulation of the physical and chemical stage of radiation action by including explicit modeling of DNA damage and repair pathways (8–10).

It is well recognized that the performance of MCTS codes critically depends upon the “quality” of the interaction cross sections that are used as input in the simulation (11). For biological media, such as liquid water, one of the most challenging issues to consider are the effects of the condensed phase, which originate from the long-range polarization of the media by the incident charged-particle field (7, 12). Neglecting such effects yields the so-called gas-phase approximation, whereby atomic (or molecular) cross sections are scaled linearly with the density of the condensed media (i.e., typically over nearly three orders of magnitude). Condensed-phase effects are known to influence both the elastic and inelastic collisions (2, 13, 14), and are therefore expected to have a noticeable effect on track structure, as confirmed by several published MCTS studies (15–17).

For inelastic scattering, condensed-phase effects are most conveniently accounted for through the dielectric function

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(18). Specifically, within the plane wave Born approximation (hereafter, simply the Born approximation) the probability of a given energy- and momentum-transfer in an inelastic collision is essentially determined by the imaginary part of the reciprocal of the dielectric function, also known as the energy-loss-function (ELF) (19). The electron gas theory provides the standard framework for calculating the dielectric function of a many-body system (20). As first shown by Lindhard (21), an analytic expression for the dielectric function of the homogeneous electron gas can be obtained within the random-phase approximation (RPA). Unfortunately, for real materials, calculating dielectric function, even at the RPA level, is a formidable task (22). The application of the Lindhard dielectric function to real materials is justified for the so-called free-electron-like materials (e.g., aluminium), which exhibit a sharp plasmon peak in their excitation spectrum that exhausts the f-sum rule at the optical limit. Despite a well-defined excitation peak near the nominal plasmon energy, biological materials (including liquid water) cannot be considered free-electron-like, since they exhibit a broadened plasmon-like peak (i.e., of a short lifetime), as well as additional excitations with non-negligible contribution to the f-sum rule. Nevertheless, electron gas theory plays a key role even for nonfree-electron-like materials by offering the theoretical framework for constructing semi-empirical dielectric functions. To that end, optical-data models of the dielectric function, first suggested by Cedric Powell of the National Institute of Standards and Technology (NIST) (23) and further developed by researchers at Oak Ridge National Laboratory (ORNL) (24, 25), provide reliable results for the inelastic scattering of low-energy electrons in arbitrary condensed media (26–29). Optical-data models are based on the simple (yet powerful) idea that the energy-dependence of the dielectric function (at the optical limit) can be obtained from experiments while the momentum-dependence can be obtained from theory. This has proved to be a very efficient method for representing the whole Bethe surface of a condensed media with only moderate computational effort (30). Obviously, fine structures in the excitation spectrum, mostly associated with low-energy discrete transitions, are often overlooked in this approach. Thus, inelastic electron-scattering calculations by optical-data models are considered most reliable for impact energies above a few tens of eV up to relativistic energies (31). In principle, the dielectric function is not applicable for nonamorphous media and must be replaced by the dielectric matrix. The latter requires band-structure calculations which are theoretically (and computationally) demanding. Such calculations for cubic ice and other biopolymers have been performed under various approximations by Zaider and colleagues (32) and implemented in their MCTS code (33).

Currently, there are three major optical-data models of the dielectric function of liquid water for use in MCTS

simulations: 1. The model of Ritchie *et al.* (25) developed at ORNL, hereafter referred to as the ORNL model; 2. The model of Dingfelder *et al.* (34) developed at the German Research Center for Environment and Health (GSF), hereafter referred to as the GSF model; and 3. The model developed by Emfietzoglou *et al.* (35), hereafter referred to as the ECN (Emfietzoglou-Cucinotta-Nikjoo) model. All three models all based on a form of the plasmon-pole approximation (36), now known as the “extended-Drude” model. The ORNL model is used in the OREC code (25), which is currently being renamed to NOREC after implementation of the NIST elastic-scattering cross sections (37). A form of the ORNL model (38–40) is implemented in the Geant4-DNA code (41) for calculating inelastic cross sections for electrons. The GSF model is used in the PARTRAC code (42) and, more recently, implemented in an unreleased version of PENELOPE (43). In the current context, the important characteristics of the ORNL and GSF models are: 1. The use of an old experimental optical data set based on reflectance measurements (44) that covers a small portion of the excitation spectrum (up to 26 eV) and is limited to zero momentum-transfer; 2. The neglect of momentum-induced lifetime-broadening effects; and 3. The dispersion of ionization energies by the impulse approximation. The ECN model, which is implemented in the KURBUC code (17), is based on a more recent experimental data set obtained by inelastic X-ray scattering (IXS) spectroscopy (45, 46). This data set covers almost the entire excitation spectrum of interest (up to ~ 150 eV) and, perhaps most importantly, extends beyond the optical limit to finite momentum transfer. Recently, a second and more accurate set of IXS measurements (47) revealed negligible discrepancies between the ECN model and the experimental data. An important feature of the ECN model is the inclusion, in a phenomenological manner, of exchange and correlation effects among the screening electrons, thus, going beyond RPA. This development, which introduces momentum-broadening and a departure from the impulse approximation, enables more accurate calculations of inelastic cross sections for low-energy electrons (48–50). Recently, an updated version of the GSF model that takes into account the IXS data (at the optical limit) has been developed by Dingfelder (51).

In the current work, inelastic cross sections beyond the Born approximation are calculated for electron energies from 10 eV to 10 keV based on the ECN model (35), and used as input in our in-house MCTS code (52) for the simulation of physical quantities relevant to the microdosimetry of low-energy electrons in liquid water. To assess the effect of new theoretical developments, we also provide a systematic comparison against track structure simulations using a simpler version of the dielectric function, based on the ORNL model (39) and the Born approximation. A description of the main theoretical developments that underline the calculation of inelastic cross sections by the ECN model is given in the next section.

MATERIALS AND METHODS

Plasmon-Pole (or Extended Drude) Approximation

The dielectric function is a key input in the study of inelastic scattering in the condensed phase because it offers a consistent description of the electronic excitation spectrum of the media through the ELF:

$$\text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] = \frac{\varepsilon_2(E, q)}{|\varepsilon(E, q)|^2}, \quad (1)$$

where $\varepsilon(E, q) = \varepsilon_1(E, q) + i\varepsilon_2(E, q)$ is the complex dielectric function, with E and q being the energy and momentum transfer, respectively. The use of q as scalar in $\varepsilon(E, q)$ implies a homogeneous and isotropic media, which is a standard approximation for liquid water (25, 29, 34).

Cross sections for individual electronic transitions are an essential input to MCTS simulations as a means of distinguishing between energy deposited locally and energy carried away by secondary electrons (e.g., δ rays). This information can be introduced into a model dielectric function as follows. In principle, the ELF may describe both single-electron excitations (interband excitations and ionizations) as well as many-electron excitations (plasmon-like collective excitations). Single-electron excitations are described by the imaginary part of the dielectric function, $\varepsilon_2(E, q)$. So, following earlier work (25, 34, 35), the assumption is made that the numerator of Eq. (1) can be partitioned as

$$\varepsilon_2(E, q) = \sum_n \varepsilon_2^{(n)}(E, q), \quad (2)$$

where the index n runs over all excitation levels and ionization shells of liquid water. However, in the condensed phase (including liquid water), the energy loss of charged particles to single-electron excitations is modified by the long-range polarization of the electronic subsystem of the media. This screening effect is represented by the denominator of Eq. (1), $|\varepsilon(E, q)|^2 = \varepsilon_1^2(E, q) + \varepsilon_2^2(E, q)$, where $\varepsilon_2(E, q)$ is given by Eq. (2) and $\varepsilon_1(E, q)$ may be determined from $\varepsilon_2(E, q)$ via the Kramers-Kronig (K-K) relationships,

$$\varepsilon_1(E, q) = 1 + \frac{1}{\pi} P \int_{-\infty}^{\infty} \varepsilon_2(W, q) (W - E)^{-1} dW, \quad (3)$$

where the symbol P stands for the Cauchy principal value. In general, finding $\varepsilon_1(E, q)$ by solving Eq. (3) may not be a straightforward process, since it will depend on the particular form of $\varepsilon_2(E, q)$. Then, Eq. (1) can be recast as

$$\text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] = \sum_n \frac{\varepsilon_2^{(n)}(E, q)}{\varepsilon_1^2(E, q) + \varepsilon_2^2(E, q)}, \quad (4)$$

It follows from the above discussion that $\varepsilon_2^{(n)}(E, q)$ is the main input for calculating inelastic cross sections for use in MCTS simulations.

A very convenient representation of the dielectric function for use in Eq. (4) is offered by the plasmon-pole approximation, which is, in its general form suitable for bound electrons, reads (36).

$$\varepsilon(E, q) = 1 + E_{pl}^2 / [E_n^2(q) - E^2 - iE\gamma_n(q)], \quad (5)$$

where $E_{pl} = 4(\pi N Z a_0^3)^{1/2}$ Ry is the nominal plasmon energy of the material with Ry the Rydberg constant, a_0 the Bohr radius, N the atomic density and Z the atomic number. The coefficients E_n and γ_n in Eq. (5) are, respectively, the excitation energy and damping (equivalent to the reciprocal lifetime of the excitation) of the n th electronic transition. Importantly, the dielectric function of Eq. (5) yields analytic forms for both its real and imaginary parts:

$$\varepsilon_1(E, q) = 1 + E_{pl}^2 [E_n^2(q) - E^2] / \{[E_n^2(q) - E^2]^2 + E^2 \gamma_n^2(q)\}, \quad (6a)$$

$$\varepsilon_2(E, q) = E_{pl}^2 E \gamma_n(q) / \{[E_n^2(q) - E^2]^2 + E^2 \gamma_n^2(q)\}. \quad (6b)$$

It is trivial to show that the plasmon-pole approximation reduces to the Drude model at the limit $q = 0$; thus, in effect, Eq. (5) represents an extended-Drude dielectric function. The extension of Eq. (5) to $q \neq 0$ depends on the particular form of the dispersion relationships, $E_n(q)$ and $\gamma_n(q)$. Since very little is known for the q -dependence of the excitation spectrum of real materials, useful information can be drawn from the known asymptotic properties at small- and large- q . For example, it can be shown that for $q \rightarrow 0$ (optical limit), the plasmon-pole approximation with $\gamma = 0$ coincides with the Lindhard dielectric function (21), whereas for $q \neq 0$ coincides with the Mermin dielectric function (53). So, to a first approximation, Eq. (5) may be extended to $q \neq 0$ using the conditions $\text{Re}[\varepsilon_L(E_{pl}(q), q)] = 0$ and $\text{Im}[\varepsilon_M(\gamma_{pl}(q), q)] = 0$, where ε_L and ε_M are the Lindhard and Mermin dielectric functions, respectively. Then, for small q , the following plasmon dispersion relationships hold (54):

$$E_{pl}^{\text{RPA}}(q) = E_{pl} + a_L(q^2/2m) \quad (7)$$

and

$$\gamma_{pl}^{\text{RPA}}(q) = \gamma_{pl}[1 + \beta_M(q^2/2m)], \quad (8)$$

where $a_L = \frac{6}{5}(E_F/E_{pl})$ and $\beta_M = \frac{16}{15}(E_F/E_{pl}^2)$ with $E_F = (\hbar^2/2m)(3\pi^2 N)^{2/3}$ the nominal Fermi energy and m the electron rest mass. On the other hand, in the vicinity of the Bethe ridge, when q is large, many-electron excitations are fully damped, and single-electron (s-e) excitations disperse as (25)

$$E_{s-e}(q) = q^2/2m, \quad (9)$$

and

$$\gamma_{s-e}(q) = \sqrt{2} q_F (q/m), \quad (10)$$

where $q_F = (2mE_F)^{1/2}$. In the absence of experimental data, the asymptotic properties expressed by Eqs. (7)–(10) offer a simple, yet physically motivated guide for extending the dielectric function beyond the optical limit.

ECN Model

For real materials, the index n in Eq. (2) includes the number of excitation levels (k) and ionization shells (j). Then, the experimental optical data for the imaginary part of the dielectric function, which cover all the outer-shell electronic transitions of liquid water, can be deconvoluted using the following expression:

$$\begin{aligned} \varepsilon_2^{\text{exp}}(E, 0) &\approx \sum_k^{\text{exc.}} \varepsilon_2^{(k)}(E, 0) + \sum_j^{\text{ioniz.}} \varepsilon_2^{(j)}(E, 0) \\ &= E_{pl}^2 \left\{ \sum_k^{\text{exc.}} \tilde{D}(E; E_k) + \sum_j^{\text{ioniz.}} D(E; E_j) \right\}, \end{aligned} \quad (11)$$

where the symbols D and $\tilde{D}$ represent the ordinary and derivative Drude functions, respectively, and coefficients f_{kij} , E_{kij} and γ_{kij} , represent the strength, energy and damping of each (k/j) electronic transition (see Appendix A). In practice, the Drude coefficients determine the height (f_{kij}), position (E_{kij}) and width (γ_{kij}) of the various excitation and ionization peaks that are used in Eq. (11) to represent the optical data. Note that the derivative Drude functions are more sharply peaked, thus, they offer a more convenient parametrization scheme for the discrete excitation peaks than the ordinary Drude functions. In the ECN model, Eq. (11) is used initially to obtain a first set of Drude coefficients by fitting the IXS data at $q = 0$ (45). Overall, five derivative Drude functions ($k = 1, 2, \dots, 5$) for the excitation levels (A¹B₁, B¹A₁, Ryd A + B, Ryd C + D, diffuse bands) and four

ordinary Drude functions ($j = 1, 2, \dots, 4$) for the outer-shells ($1b_1, 3a_1, 1b_2, 2a_1$) are used. Subsequently, $\varepsilon_1(E, 0)$ and $\text{Im}[-1/\varepsilon(E, 0)]$ are calculated from Eqs. (3) and (4), respectively, and tested against the corresponding experimental optical data. The final set of Drude coefficients is obtained after ensuring that the f-sum rules for $\varepsilon_2(E, 0)$ and $\text{Im}[-1/\varepsilon(E, 0)]$ are fulfilled to better than 1%. A newer and more accurate set of IXS data at $q = 0$ published recently (47) has further confirmed the accuracy of the ECN model. As noted in (34), Eq. (11) ignores the various threshold energies associated with the excitation levels and the ionization shells, since it yields finite values for $\varepsilon_2(E, 0)$ for any non-zero value of E . To circumvent this problem, a post-processing is performed by recasting Eq. (11) as:

$$\varepsilon_2(E, q = 0) = E_{pl}^2 \left\{ \sum_k^{\text{exc.}} \tilde{D}_{\text{mod}}(E; E_k) + \sum_j^{\text{ioniz.}} D_{\text{mod}}(E; E_j) \right\}, \quad (12)$$

where the modified Drude functions have the form

$$\tilde{D}_{\text{mod}}(E; E_k) = \left\{ \sum_k^{\text{exc.}} [\tilde{D}(E; E_k) + F_k(E)] \right\} \Theta(E - B_k), \quad (13a)$$

$$D_{\text{mod}}(E; E_j) = \left\{ \sum_j^{\text{ioniz.}} [D(E; E_j) - D(E; B_j) \exp(B_j - E) + F_j(E)] \right\} \Theta(E - B_j), \quad (13b)$$

with $\Theta(\dots)$ the unit step function, B_{kj} the threshold energy for excitation (k) or ionization (j), $D(E; B_j) \exp(B_j - E)$ an exponential smoothing function for ionizations and $F_{kj}(E)$ contributions due to the smoothing and truncation of higher-energy-level Drude functions. The contributions $F_{kj}(E)$ are calculated analytically by a redistribution of the original Drude functions (see Appendix B). Importantly, the post-processing (truncation, smoothing, redistribution) of the Drude functions, as described by Eq. (13a, b), conserves the f-sum rule since

$$\begin{aligned} \int_0^\infty \left\{ \sum_k^{\text{exc.}} \tilde{D}(E; E_k) + \sum_j^{\text{ioniz.}} D(E; E_j) \right\} E dE \\ = \int_0^\infty \left\{ \sum_k^{\text{exc.}} \tilde{D}_{\text{mod}}(E; E_k) + \sum_j^{\text{ioniz.}} D_{\text{mod}}(E; E_j) \right\} E dE. \end{aligned} \quad (14)$$

Thus, $\varepsilon_1(E, 0)$ can still be calculated from the original Drude functions using the expression:

$$\varepsilon_1(E, q = 0) = 1 + E_{pl}^2 \left\{ \sum_k^{\text{exc.}} \tilde{D}_{K-K}(E; E_k) + \sum_j^{\text{ioniz.}} D_{K-K}(E; E_j) \right\}, \quad (15)$$

where $\tilde{D}_{K-K}$ and D_{K-K} are the Kramers-Kronig pairs of the derivative and ordinary Drude functions, respectively, obtained from Eq. (3) (see Appendix A). Importantly, since both $\tilde{D}_{K-K}$ and D_{K-K} have a simple analytic form, the calculation of $\varepsilon_1(E, 0)$, via Eq. (15), is straightforward. This is an important outcome since the calculation of the K-K integral, Eq. (3), of truncated Drude functions is complicated. Thus, the post-processing algorithm enables the inclusion of excitation and ionization cut-off energies in $\varepsilon_2(E, 0)$ while retaining a rather simple and fully analytic form for $\varepsilon_1(E, 0)$.

Another important feature of the ECN model is that the dispersion relationships are determined empirically from the IXS data at $q \neq 0$ (46). According to the general expectation (and earlier studies), different dispersion relationships are used for the excitation levels and the ionization shells. Specifically, in the ECN model, ionizations are dispersed according to the following relationship (q in atomic

units):

$$E_j(q) = E_j + a_{\text{emp}}(q) \text{Ry} q^2, \quad (16)$$

where $a_{\text{emp}}(q) = 1 - \exp(-cq^d)$ with the empirical coefficients $c = 1.5$ and $d = 0.4$ deduced from the fit to the IXS data. Note that the form of Eq. (16) is similar to the RPA dispersion relationship, Eq. (7). However, contrary to Eq. (7), which is valid only at small q , the form of $a_{\text{emp}}(q)$ ensures the correct limiting behavior for $E_j(q)$ at both small and large q . Specifically, for $q \rightarrow 0$ we get $a_{\text{emp}} = 0$ and $E_j(q) = E_j$, and for $q \rightarrow \infty$ we get $a_{\text{emp}} = 1$ and $E_j(q) = \text{Ry} q^2$, which coincides with Eq. (9) when q is expressed in the same units. Moreover, since the Lindhard dispersion coefficient is about unity for liquid water ($a_L \approx 1$), the effect of $a_{\text{emp}}(q)$ for not too large q is to reduce $E_j(q)$ from the corresponding RPA value of Eq. (7). This behavior resembles the effect of exchange and correlation (XC) in many-electron systems which, to second order in q , diminishes the RPA dispersion coefficient (55). On the other hand, the excitation levels are assumed to be nondispersive, i.e., $E_k(q) = E_k$ (for all k), as it is the case for most interband excitations. In the ECN model, lifetime broadening is explicitly considered through a linear-quadratic dispersion relationship (q in atomic units)

$$\gamma_{kj}(q) = \gamma_{kj} + \beta_1(\text{Ry} q) + \beta_2(\text{Ry} q^2), \quad (17)$$

with the empirical coefficients $\beta_1 = 0.735$ and $\beta_2 = 0.441$ deduced from the fit to the IXS data. The form of Eq. (17) takes into account the quadratic dispersion of many-electron excitations at small q [see Eq. (8)] and the linear dispersion of single-electron excitations at large q [see Eq. (10)]. Calculations of the momentum-dependent ELF by Eqs. (16) and (17) have been shown to be equivalent to the time-dependent local-density approximation (55). Finally, the dispersion relationships for the oscillator-strength coefficients, $f_{kj}(q)$, in the ECN model are identical to those used in the ORNL and GSF models.

ORNL Version

The ECN model reduces to the ORNL version of the dielectric function if: 1. the Drude coefficients of the IXS optical data are replaced by those corresponding to the old (reflectance) optical data set of Heller *et al.* (44) as suggested by Emfietzoglou and Nikjoo (39); and 2. the dispersion relationships Eqs. (16) and (17) are simplified by taking $a_{\text{emp}}(q) = 1$ and $\beta_{1,2} = 0$, respectively. The first condition ($a_{\text{emp}} = 1$) leads to the impulse approximation for the ionization energies (q in atomic units):

$$E_j(q) = E_j + \text{Ry} q^2, \quad (18)$$

which properly yields the Bethe ridge limit of Eq. (9), since $E_j(q) \rightarrow \text{Ry} q^2$ for $E_j \ll \text{Ry} q^2$. However, Eq. (18) is expected to be less accurate than Eq. (16) for not too large values of q , where many electron excitations and other many body effects set in. The second condition ($\beta_{1,2} = 0$) leads to the neglect of momentum-induced lifetime-broadening effects by transforming Eq. (17) to $\gamma_{kj}(q) = \gamma_{kj}$ (for all k/j). The impulse approximation and the lack of broadening adopted by the ORNL and GSF models have been shown to yield a Bethe ridge that is too narrow and displaced to higher energy losses compared to the IXS data.

It must be made clear that what we referred to in this article as the ‘‘ORNL version’’ of the dielectric function is not identical to the original ORNL model, but shares with the latter its two most important ingredients, namely, the old (reflectance) optical data and the form of the dispersion relationships. The current ORNL version differs from the original ORNL model in the details of the parameterization of the optical data; for example, whereas in the original ORNL model the partitioning algorithm of the dielectric function includes a plasmon channel, this channel is not considered in the current ORNL version.

Other Models

As discussed above, the three most popular optical-data models for the dielectric function of liquid water (ORNL, GSF, ECN) are Drude-based models. Other optical-data models, most notably the Penn model (56) and the MELF model (57), have been extensively used for calculating inelastic mean free paths and stopping powers for charged-particles in arbitrary solids, including several biological materials and liquid water (58–61). An attractive feature of the Penn and MELF models is the use of the more sophisticated Lindhard (21) and Mermin (53) dielectric functions, respectively, where the q -dependence is built in. Thus, the extension of the optical data to $q \neq 0$ in both the Penn and MELF models is done automatically without the need for adopting a particular dispersion relationship, as is the case with the Drude-based models. Nevertheless, the Penn and MELF models have not been very popular for MCTS simulations because information on partial cross sections, associated with the individual ionization shells and excitation levels, cannot be readily deduced. The source of this limitation is that both models are essentially parameterizations of the ELF and not the dielectric function, so further approximations (and uncertainties) are introduced when the ELF is partitioned to the various energy-loss channels.

Born Approximation

Within the Born approximation, the ELF is the only nontrivial input in calculating inelastic cross sections. Specifically, the double-differential inverse inelastic mean free path takes the form:

$$\frac{d^2\Lambda_{\text{Born}}(E, q; T)}{dE dq} = \frac{1}{\pi a_0 T} \frac{1}{q} \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right], \quad (19)$$

where T is the electron (kinetic) energy, and Λ is the inverse inelastic mean free path or macroscopic inelastic cross section (in units of reciprocal length). The latter is related to the more conventional microscopic cross section through $\Lambda = N_{\text{targets}}\sigma$, where N_{targets} is the number of targets (electrons, atoms, molecules) per unit volume. In principle, Eq. (19) can be used in MCTS simulations to determine both the energy-loss E and the momentum-transfer q (or scattering angle) in an inelastic collision. However, sampling from two-dimensional tables of ELF is impractical. Therefore, following common practice, the ELF is used in our MCTS code to sample the energy-loss associated with the various excitations and (outer-shell) ionizations through the corresponding partial single-differential inverse inelastic mean free paths:

$$\frac{d\Lambda_{\text{Born}}^{(k/j)}(E; T)}{dE} = \frac{1}{\pi a_0 T} \int_{q_{\min}}^{q_{\max}} \frac{\varepsilon_2^{(k/j)}(E, q)}{|\varepsilon(E, q)|^2} \frac{dq}{q}, \quad (20)$$

with the limits of integration being $q_{\min} = \sqrt{2m}(\sqrt{T} - \sqrt{T-E})$ and $q_{\max} = \sqrt{2m}(\sqrt{T} + \sqrt{T-E})$. The dielectric function in Eq. (20) is here determined by the ECN model. Integration of Eq. (20) over E yields the inverse inelastic mean free path associated with each excitation level k and ionization shell j :

$$\Lambda_{\text{Born}}^{(k/j)}(T) = \frac{1}{\pi a_0 T} \int_{E_{\min}^{(k/j)}}^{E_{\max}^{(k/j)}} dE \int_{q_{\min}}^{q_{\max}} \frac{\varepsilon_2^{(k/j)}(E, q)}{|\varepsilon(E, q)|^2} \frac{dq}{q}, \quad (21)$$

where the limits of integration are set as $E_{\min}^{(k/j)} = B_{k/j}$, $E_{\max}^{(k)} = T$, and $E_{\max}^{(j)} = \frac{1}{2}(T + B_j)$; the factor $\frac{1}{2}$ in $E_{\max}^{(j)}$ accounts for the indistinguishability between the primary and secondary electron).

For collisions associated with large E (and q), such as those associated with inner-shell ionizations, the screening factor tends to unity, $|\varepsilon(E, q)| \rightarrow 1$, and Eq. (4) reduces to Eq. (2), that is, $\text{Im}[-1/\varepsilon(E, q)] = \varepsilon_2(E, q)$, which is also the condition that holds in the gas phase. Thus, for the oxygen K-shell, which has a binding energy of about 540 eV and, therefore, a minimum contribution (<1%) to inelastic cross sections at the current energy range, Eqs. (20) and (21)

are replaced by the corresponding formulas of the binary-encounter-approximation-with-exchange (BEAX) (62). The BEAX model is known to provide reasonably accurate results for inner-shell ionizations of atoms and molecules by electron impact (63).

Born Corrections

The Born cross section Eqs. (19)–(21) are based on first-order perturbation theory, which is valid only for fast projectiles with velocity that (much) exceeds the velocity of target electrons. In addition, the Born approximation neglects exchange effects in the interaction of the incident electron with the target electrons. It is generally recognized that for electrons in the sub-keV energy range, corrections to the Born approximation due to the above deficiencies are needed (64). So far, the corrections to the Born approximation used in MCTS simulations are either empirical or heuristic (52). In the current work, such corrections are calculated using the ECN model as follows. The single-differential inverse inelastic mean free path is expressed by the following sum of terms:

$$\frac{d\Lambda^{(k/j)}(E; T)}{dE} = \frac{d\Lambda_{\text{Born}}^{(k/j)}(E; T)}{dE} + \frac{d\Lambda_{\text{2nd-order}}^{(k/j)}(E; T)}{dE} + \frac{d\Lambda_{\text{Ex}}^{(j)}(E; T)}{dE}. \quad (22)$$

The Born term is given by Eq. (20). The second-order term is calculated from Ashley's soft-collision approximation, which is proportional to the optical limit of the ELF (65),

$$\frac{d\Lambda_{\text{2nd}}^{(k/j)}(E; T)}{dE} = -\frac{2}{\pi a_0 T} \frac{\varepsilon_2^{(k/j)}(E, 0)}{|\varepsilon(E, 0)|^2} L(E), \quad (23)$$

with $L(E)$ a tabulated function. The exchange term is calculated in analogy to the nonrelativistic Moller formula by setting the direct term equal to the sum of the Born and second-order terms:

$$\begin{aligned} \frac{d\Lambda_{\text{Ex}}^{(j)}(E; T)}{dE} &= \frac{d\Lambda_{\text{Born+2ndOrder}}^{(j)}(T - E + B_j; T)}{dE} \\ &\quad - \sqrt{\frac{d\Lambda_{\text{Born+2ndOrder}}^{(j)}(E; T)}{dE} \times \frac{d\Lambda_{\text{Born+2ndOrder}}^{(j)}(T - E + B_j; T)}{dE}}. \end{aligned} \quad (24)$$

As shown in (52), Ashley's soft-collision approximation, Eq. (23), becomes gradually invalid as the electron energy falls below 100 eV. So, for $T < 100$ eV, the second-order term in Eq. (22) is replaced by the Coulomb-field-corrected Born term (66). This, essentially, amounts to the kinematic transformation $\Lambda_{\text{Born}}^{(k/j)}(T \rightarrow T')$, with $T' = T + 2E_k$ in the case of excitation, and $T' = T + B_j + U_j$ in the case of ionization (U_j being the average kinetic energy of the electron at the j -th shell).

It should be emphasized that introducing corrections to the Born approximation is far from trivial, especially for condensed media. The above scheme, Eqs. (22)–(24), provides a reasonable, physically-motivated scheme for going beyond the Born approximation, considering the overall requirements (and limitations) of MCTS simulations. To that end, it is somewhat comforting to note that all terms in Eq. (22) use as input the dielectric function of the media (here the ECN model) and, as such, they can be considered "material-specific". A review of available schemes for introducing, in a practical manner suitable for MCTS simulations, Born corrections to inelastic scattering is provided in (52). More recently, considerations of exchange and correlation effects to inelastic electron scattering within the dielectric function approach have been presented at a more sophisticated level (55, 67, 68).

Angular Deflections

The scattering angle after an inelastic or elastic collision is simulated as in our earlier version of the code (52). For electrons

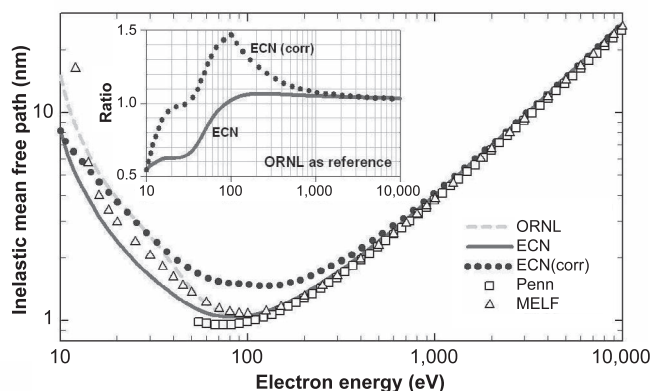


FIG. 1. Comparison of calculated inelastic mean free paths (IMFP) as a function of electron energy. Dashed-line: calculations using the ORNL version of the dielectric function and the Born approximation; solid line: calculations using the ECN model of the dielectric function and the Born approximation; dotted-line: calculations using the ECN model of the dielectric function with Born corrections (see text); squares: calculations using the MELF model and the Born-Ochkur exchange approximation from (60); triangles: calculations using the Penn model and the Born approximation from (61). The inset shows the IMFP ratio between the following models: solid line: $ECN \div ORNL$; dotted-line: $ECN(corr) \div ECN$.

above 200 eV, the screened-Rutherford formula with the empirical screening parameter suggested in (69) is used. Below 200 eV the screened-Rutherford differential cross section is replaced by the Brenner-Zaider (70) empirical parameterization. For electrons above 100 eV, the angular distributions of the scattered and secondary electrons after inelastic collision (ionization) are obtained from the (nonrelativistic) kinematic expressions of binary collisions. Below 100 eV, we use the prescription of Grosswendt and Waibel (71), which is based on the experimental data of Opel *et al.* (72). Finally, following standard practice, it is assumed that excitations do not cause any angular deflection, which, even if not entirely correct, is justified in practice by the small (on average) momentum-transfer in such collisions.

RESULTS AND DISCUSSION

The MCTS results presented in this section are based on one-million simulated tracks for electron energies below 5 keV, and 500,000 simulated tracks for electron energies above 5 keV. The cut-off transport energy was set at 1 Ry ($= 13.606$ eV) where electronic excitations (and ionizations), which are the main interest in this work, are still the dominant energy-loss channel. Electrons with energy below 1 Ry are “killed” and their energy is assumed to be deposited at the spot. In all simulations, the elastic model (as described above) was kept the same.

At low-electron energies and very small target volumes, the inelastic mean free path (IMFP), Λ^{-1} , is one of the most important quantities for track-structure studies. Therefore, before the presentation and discussion of the MCTS results, we highlight in Fig. 1 the IMFP as a function of electron energy from 10 keV down to 10 eV. Comparison is made between the IMFP calculated by the ECN model and the

ORNL version in the Born approximation, as well as by the ECN model with Born corrections. Differences between these model calculations using the ORNL results as baseline, i.e., the ratios $ECN \div ORNL$ and $ECN(corr) \div ORNL$, are shown in the inset. Also included are the most recent calculations of IMFP for liquid water with the Penn (61) and MELF models (60). Clearly, the different model calculations are in good agreement above a few hundred electron-volts, but sizeable differences are observed at lower energies. Specifically, the ECN model yields smaller IMFP compared to the ORNL version below 100 eV due to the inclusion of non-RPA effects in the dispersion relationships, which yield a broadened and red-shifted ELF (2, 49, 50). In contrast, Born corrections to the ECN model enhance the IMFP because of a smaller interaction probability, in agreement with the general expectation and other model calculations (51, 53). Overall, the calculations by the Penn and MELF models appear to be closer to the ECN model. The Penn calculations do not include any Born corrections, which, in addition to the Landau damping mechanism in the Lindhard dielectric function, result in the smallest IMFPs at low energies. The MELF model includes, due the properties of the Mermin dielectric function, momentum-broadening effects more consistently than the Penn model, which should have resulted in even smaller IMFPs. However, the calculations by the MELF model in (60) include exchange corrections through the Born-Ochkur approximations. This approximation yields a much smaller correction to the IMFP than the current ones (39, 55). Thus, the MELF calculations are somewhat between the current ECN and the ECN(corr) models.

One of the most basic electron track parameters is the penetration distance in the media, which, due to the three-dimensional nature of the track, can take several forms. In Fig. 2 we present MCTS calculations of two such penetration parameters, namely, the path length (Fig. 2A) and the absorption distance (Fig. 2B), as a function of primary electron energy. The path length is defined as the average value of the total distance traveled by the primary electron until its kinetic energy falls below the transport cut-off energy (here 1 Ry). Thus, if we designate by $\vec{r}_i$ the vectors that connect the energy-transfer points of the primary electron with Cartesian coordinates $(x_{i-1}, y_{i-1}, z_{i-1})$ and (x_i, y_i, z_i) , then the average value of $r_{total} = \sum_{i=1}^n |\vec{r}_i|$ defines the path length. The path length is closely related (but not exactly equal) to the continuous-slowing-down-approximation range calculated from the integral of the reciprocal of the stopping power. On the other hand, the absorption distance is the average length of the vector that connects the final energy-transfer point (i.e., absorption point) of the primary electron with its starting point. If we set the track starting point at $(x_0, y_0, z_0) = (0, 0, 0)$ and the absorption point of the primary electron at $(x_{abs}, y_{abs}, z_{abs})$, then the average value of $r_{abs} = \sqrt{x_{abs}^2 + y_{abs}^2 + z_{abs}^2}$ defines the absorption distance. Since electrons follow tortuous paths due to significant angular deflections, r_{total} is always

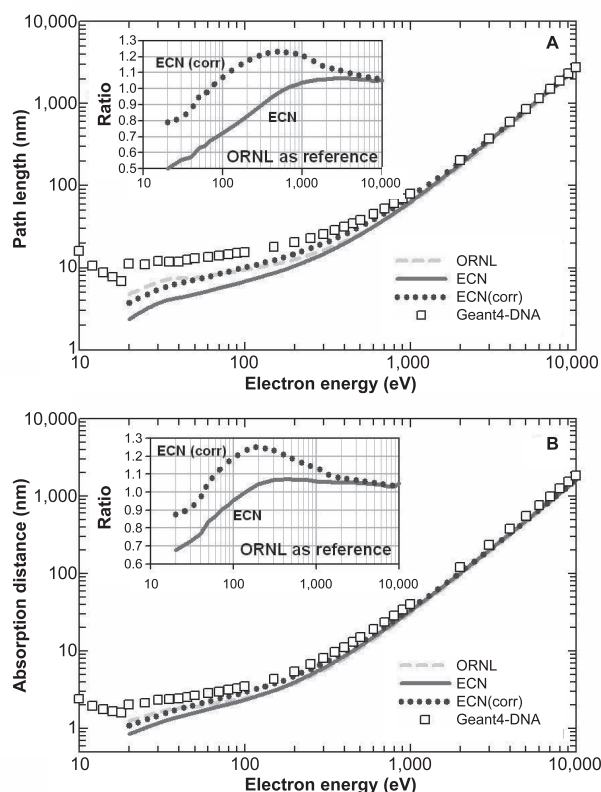


FIG. 2. Monte Carlo track structure simulations of the path length (panel A) and absorption distance (panel B) as a function of incident electron energy. Dashed line: simulations using the ORNL version of the dielectric function and the Born approximation; solid line: simulations using the ECN model of the dielectric function and the Born approximation; dotted-line: simulations using the ECN model of the dielectric function with Born corrections (see text); squares: Geant4-DNA simulations from (73). Insets show the path length (panel A) and absorption distance (panel B) ratios between the following models: Solid line: $ECN \div ORNL$; dotted line: $ECN(corr) \div ECN$.

larger than r_{abs} . Note that the above definitions concern solely the primary electron, so transport of secondary electrons is not required for calculating either r_{total} or r_{abs} . In Fig. 2 we compare results of simulations of r_{total} (Fig. 2A) and r_{abs} (Fig. 2B) based on the dielectric function of either the ECN model or the ORNL version. For the ECN model, results with and without Born corrections are shown. In addition, recent Geant4-DNA simulation results (Geant4 version 10.2) are also shown for comparison (73). Overall, differences are small for primary electron energies above a few keV, with the ECN model yielding slightly larger (up to 10%) values of r_{total} and r_{abs} compared to the ORNL version. At sub-keV energies, however, discrepancies become larger, at the level of 20–50%. Specifically, the ECN model yields smaller r_{total} and r_{abs} compared to the ORNL version, due to the shorter IMFP at this energy range (see Fig. 1). As expected, including Born corrections enables the electrons to travel further in the media due to

an increase of the IMFP (see Fig. 1), thus, the ECN(corr) model yields larger r_{total} and r_{abs} than the (uncorrected) ECN model.

Overall, the Geant4-DNA simulation results compare fairly well with the current work, especially for electron energies above a few hundred eV. At lower energies, discrepancies increase because the details of the inelastic-scattering model become crucial (the elastic-scattering model is the same in both studies). The Geant4-DNA simulations shown in Fig. 2 are based on the latest model of the dielectric function implemented in the Geant4 code (40). This model is similar to our ORNL version but with the addition of Born corrections. Thus, the somewhat larger values of Geant4-DNA for r_{total} and r_{abs} , compared to our ORNL version, must be attributed to the larger IMFPs (due to Born corrections) as well as to the lower simulation cut-off energy used in the Geant4-DNA simulations (10 eV or lower).

A very useful distribution in dosimetry is the dose-point-kernel (DPK), which is generally defined as $DPK = 4\pi r^2 \rho D(r)$ with ρ the mass density of the media, r the radial distance from the emission point and $D(r)$ the dose at distance r . Thus, the DPK is the average energy that is deposited in concentric spherical shells of radius r and thickness dr around an emission point. In Fig. 3, normalized DPKs are shown, which represent the fraction (%) of the primary electron energy that is deposited between r and $r + dr$. Results are shown for four primary electron energies: 50, 250, 500 and 1,000 eV. For not too low-electron energies, the ECN model yields a more diffused DPK compared to the ORNL version, due to the higher penetration ability of the primary electron for energies above 200 eV (Fig. 2B, see inset). However, for electrons below approximately 100 eV the situation is reversed and the ECN model yields a less diffused DPK, in agreement with the penetration results (Fig. 2B, see inset). For all electron energies, including Born corrections to the ECN model, yields are a more diffused DPK due to a larger penetration ability (Fig. 2).

A mechanistic understanding of radiation action in cells requires, as a first step, knowledge of energy deposition in subcellular structures such as, for example, the DNA, the nucleosome and the chromatin. The critical dimensions of these structures range from approximately 2–30 nm. For such small targets the stochastic fluctuations of energy deposition are significant (except for extremely high doses), so neither the dose nor the LET provides a suitable physical description for biophysical models of radiation action. In such cases, microdosimetric quantities, like the specific energy (z) and the lineal energy (y), are more suitable (73). Figures 4 and 5 show MCTS simulation results for the frequency- and dose-mean lineal energy, denoted as y_F and y_D , respectively, for target spheres with diameter of 2, 5, 10, 30 nm. Of note, y_F is the microdosimetry analogue of LET and y_D is the microdosimetry analogue of dose-mean LET, which is an important description of radiation quality and an essential parameter in some biophysical models of radiation

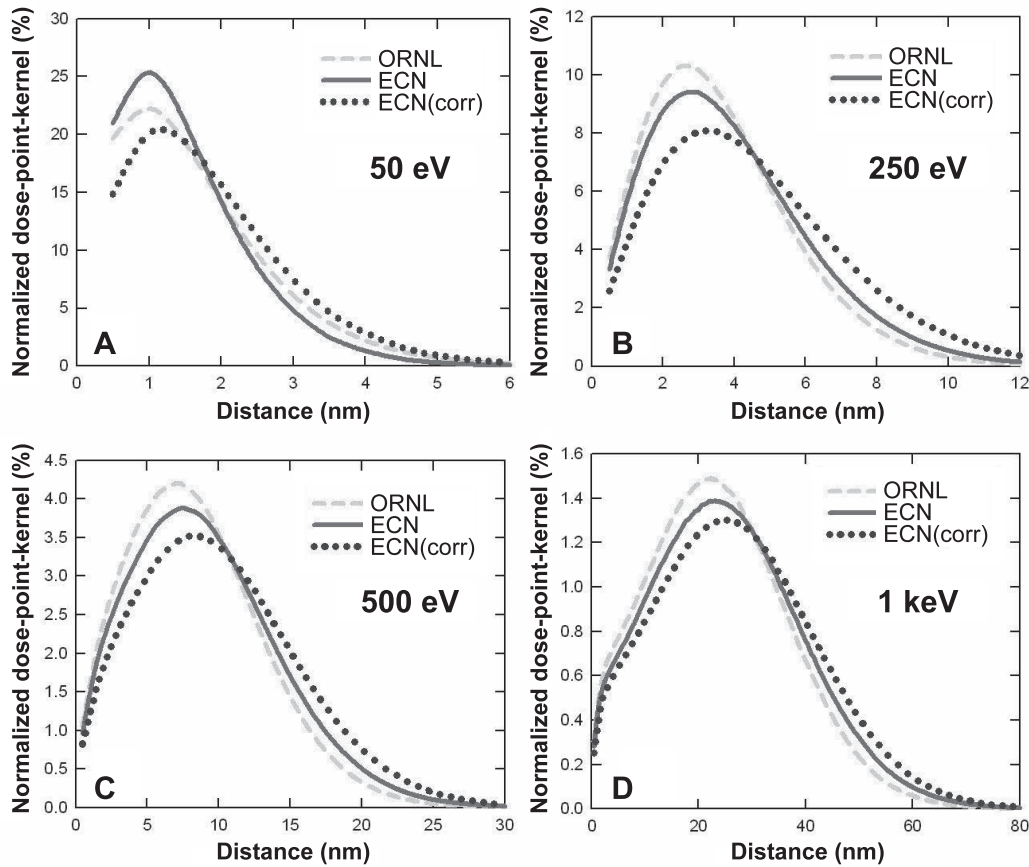


FIG. 3. Monte Carlo track structure simulations of the (normalized) dose-point-kernel for electrons with initial energy 50 eV (panel A), 250 eV (panel B), 500 eV (panel C) and 1,000 eV (panel D). Dashed line: simulations using the ORNL version of the dielectric function and the Born approximation; solid line: simulations using the ECN model of the dielectric function and the Born approximation; dotted-line: simulations using the ECN model of the dielectric function with Born corrections (see text).

action (74). The scoring of energy deposition inside the target sphere is based on the concept of the associated volume of the track. In brief, we place within a distance r from a randomly selected transfer point, a sphere of radius r and calculate the total energy deposition in this sphere. To avoid the bias of sampling regions of the track with high density of transfer points, the value of energy deposition thus obtained is scored in the spectrum with a weight inversely proportional to the number of transfer points in the sphere (75). Results are shown for many primary electron energies from 50 eV to 10 keV. The ECN model leads to higher energy deposition than the ORNL version for the smaller volumes for all primary energies. This is because the very-low-energy secondaries (track-ends) have shorter IMFPs in the ECN model, resulting in a more dense pattern of energy deposition, especially near the end of the track. Differences in lineal energy due to Born corrections are more significant and they are also noticeable even for the larger spheres. As shown in Fig. 1, Born corrections lead to larger IMFPs, thereby decreasing the energy deposition in the target volumes.

Another fundamental description of track structure is the proximity function, which is defined as the mean energy deposited to a spherical shell of radius r and thickness dr centered at a randomly chosen energy-transfer point on the track. The proximity function is defined as $t(r) = d \bar{t}(r) / dr$ with $\bar{t}(r) = \sum_{i,j} \varepsilon_i \varepsilon_j / \sum_i \varepsilon_i$, where the summations run over all energy-transfer points of the track (including $i = j$), ε_{ij} is the energy-transfer at the point i/j and the distance between two energy-transfer points, i/j , is smaller than r . Evidently, the proximity function provides a measure of the clustering properties of the track. It can be shown that from the proximity function the dose-mean lineal energy can be computed for any specified target volume (76). Figure 6 shows proximity functions for primary electron energies: 50, 100, 250, 500, 1,000 and 5,000 eV. Sizeable differences are observed between the ECN model and its ORNL version for distances up to a few nanometers. The effect of Born corrections in the ECN model is also most significant up to a few nanometers. Interestingly, the above differences persist over the entire energy range studied. To a good

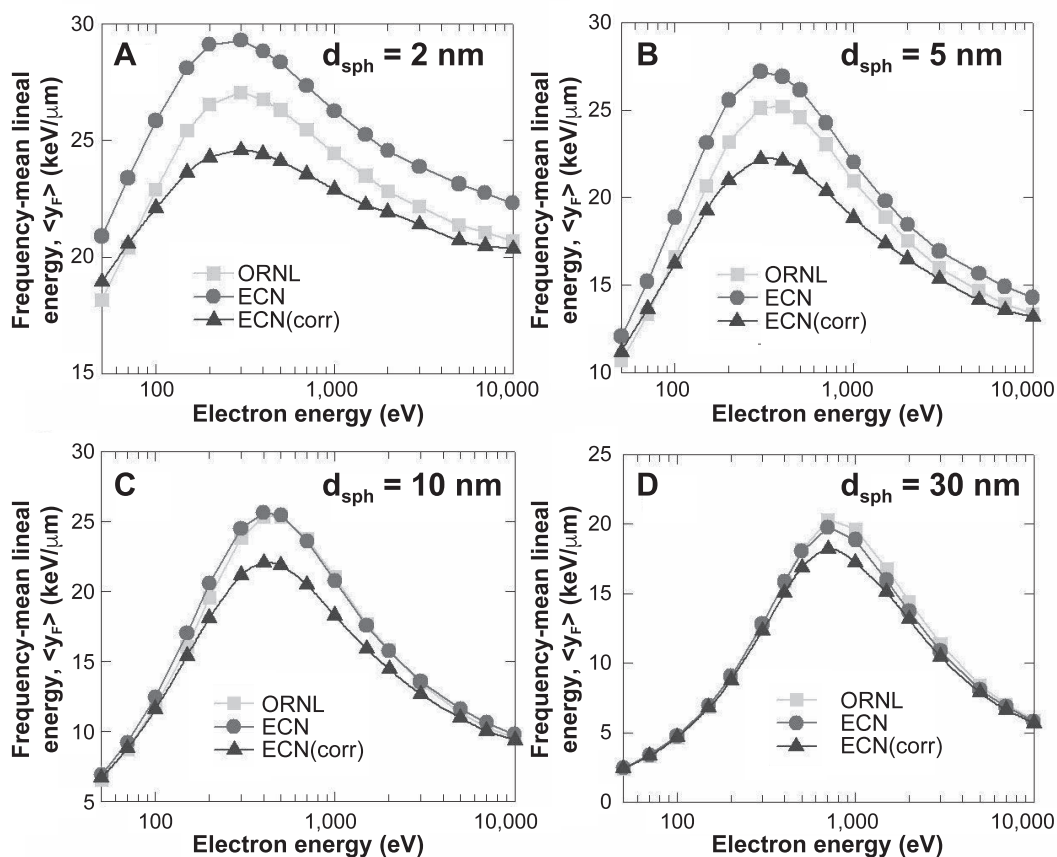


FIG. 4. Monte Carlo track structure simulations of the frequency-mean lineal energy, $\bar{y}_F$, as a function of incident electron energy for target spheres with 2 nm (panel A), 5 nm (panel B), 10 nm (panel C) and 30 nm (panel D) diameter. Squares: simulations using the ORNL version of the dielectric function and the Born approximation; circles: simulations using the ECN model of the dielectric function and the Born approximation; triangles: simulations using the ECN model of the dielectric function with Born corrections (see text).

approximation, and in all cases, no differences are observed for distances beyond 10 nm.

It is worth noting that, in some cases, the ECN(corr) results are, overall, closer to the (uncorrected) ORNL version than the (uncorrected) ECN model. This is fortuitous and occurs because corrections to RPA and corrections to the Born approximation affect the IMFP at opposite directions. Specifically, corrections to RPA (e.g., broadening) reduce the IMFP, whereas corrections to the Born approximation increase the IMFP. This is evident in Fig. 1 (inset) where the ECN(corr) model is initially higher (up to factor of 1.5 at 100 eV) from the ORNL version but, eventually, becomes smaller (up to a factor of one-half at 20 eV). Thus, when the IMFP is averaged over a sufficient energy range or track segment, the non-RPA and non-Born effects included in the ECN(corr) model tend to cancel out to produce results that, in some cases, resemble those of the (uncorrected) ORNL version.

Although the simulation results presented have a statistical uncertainty that is, overall, smaller than 1%, the systematic uncertainty cannot be easily determined. It is

generally recognized that the physics models used as input in MCTS codes have a much larger uncertainty margin compared to those used in macroscopic (condensed-history) MC codes. In addition to the fundamental question on the validity of the (classical) trajectory simulation for sub-keV electron transport [recently challenged in (77) and defended in (78)], the main uncertainty in the current simulations comes from the choice of the elastic and inelastic-scattering models. The physics of low-energy electron transport has been recognized as one of the main challenges in MCTS studies (2, 79). The screened Rutherford elastic-scattering model used in the current work is a high-energy (Born approximation) model empirically adjusted to low-energy gaseous water data. Therefore, it may not be accurate below a few hundred electron-volts where condensed-phase effects are most pronounced. Although more elaborate quantum mechanical models are available based on partial wave analysis, it is not clear if they could accurately incorporate the condensed-phase effects (80). For example, recent partial wave calculations of the elastic-scattering cross section of liquid water predict (only) a 10–15% reduction

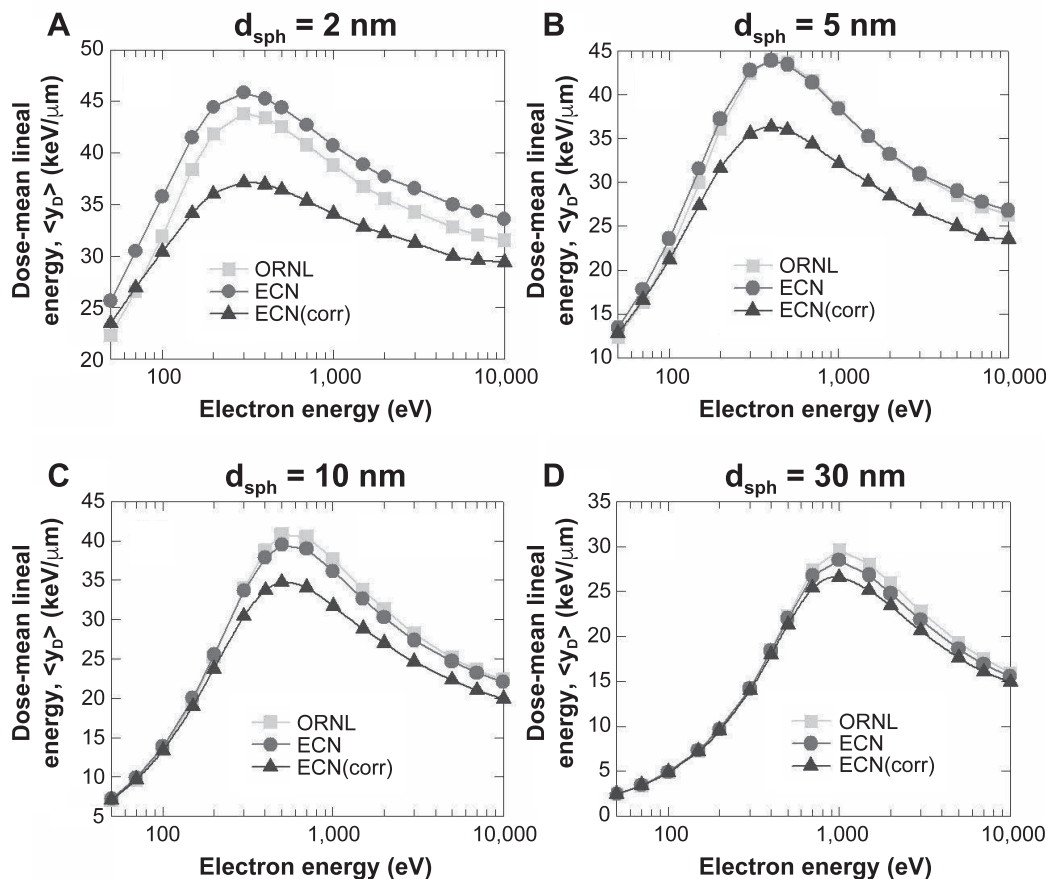


FIG. 5. Same as in Fig. 4 for the dose-mean lineal energy, $\bar{y}_D$.

from the corresponding gas-phase data (81), whereas reduction of nearly two orders of magnitude is predicted from elaborate measurements in amorphous ice (82). In principle, a reduction of the elastic-scattering cross section (as expected in the condensed phase) should lead to more rectilinear tracks. Concerning the inelastic-scattering model, one can recognize at least three sources of uncertainty in the current model (50). The first source of uncertainty concerns the optical data. The IXS data at $q = 0$ (45) have been recently confirmed (to within $\sim 5\%$) by a second and more accurate set of measurements (47). Any remaining uncertainty arising from the high energy-loss tail (above 100 eV) should have a very small overall effect in the current calculations. Similarly, any uncertainties related to the nonuniqueness of the partitioning of the optical data to the individual energy-loss channels should not be significant for dosimetry (unless perhaps one is interested in simulating the subsequent radiation chemistry). *Ab initio* calculations of optical data are perhaps the only way to determine the contribution of individual channels in a (more) realistic manner (83). The second source of uncertainty concerns the extension of the optical data to $q \neq 0$. The ECN model is based on a reasonably faithful parameterization of the IXS data at finite q (46) and, as such, it should not contain uncertainties much beyond those associated with the

experimental data (reported to be insignificant). Obviously, a second set of measurements at $q \neq 0$ is highly desirable to confirm the efficiency of the ECN model beyond the optical limit. The third source of uncertainty concerns the corrections to the Born approximation. Recently published theoretical studies (50, 55) on the effects of exchange (and correlation) to inelastic scattering within the dielectric function approach have revealed a sizeable increase of the IMFP. An effect of a similar magnitude is also predicted by the current model. The recently published experiments in amorphous ice, however, seem to point to even larger IMFPs below 100 eV (82). This is an area that deserves further work and where more experimental data for direct comparison with theoretical cross sections are urgently needed. Based on the above discussion, we expect that future improvements in the current models for elastic and inelastic scattering will, likely, tend to increase the corresponding mean free paths. This in turn will result in more diffused electron tracks having a less dense pattern of energy-transfer points.

CONCLUSION

Monte Carlo track structure simulations of physical quantities relevant to the microdosimetry of low-energy

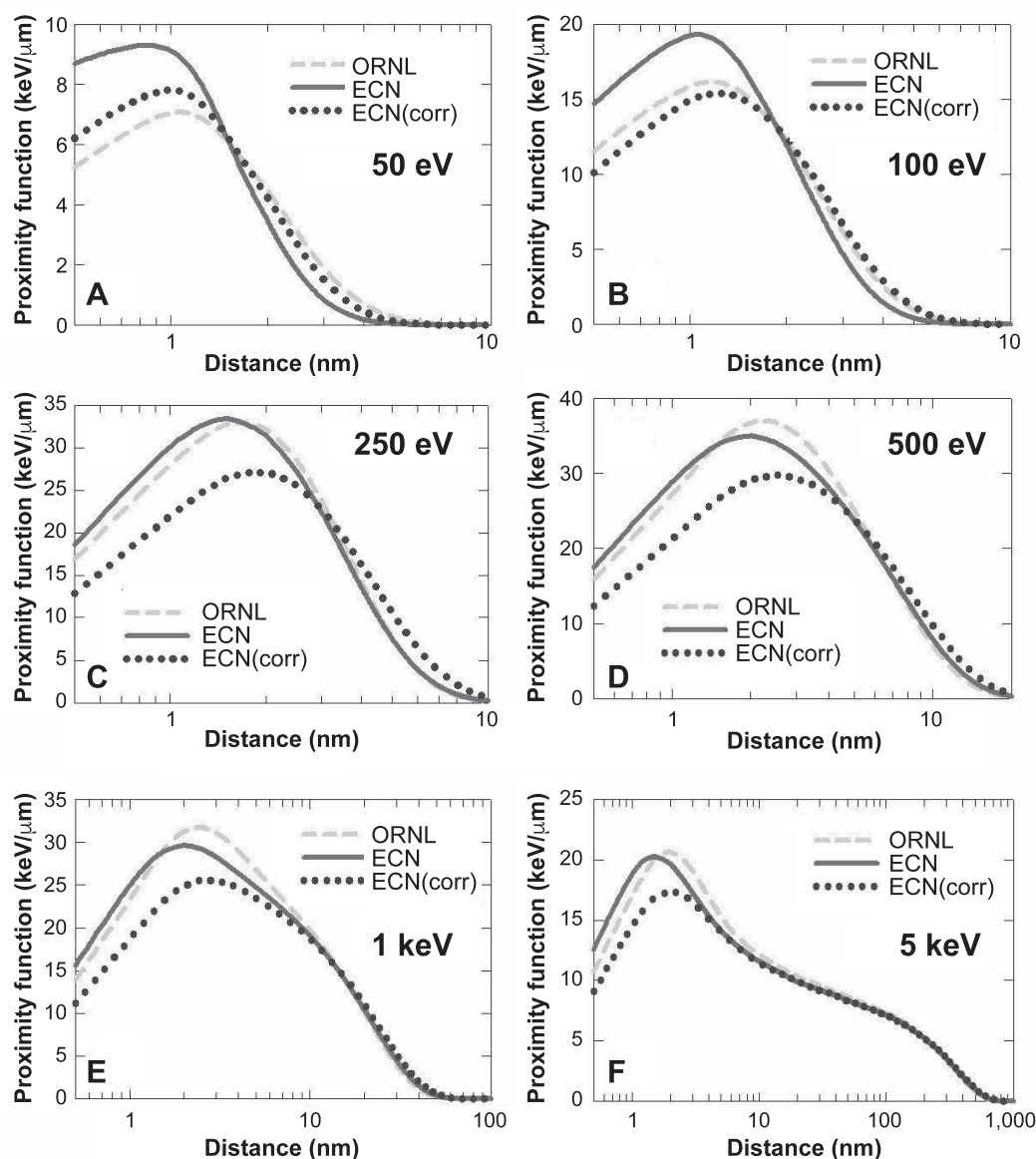


FIG. 6. Monte Carlo track structure simulations of the proximity functions, $t(r)$, for electrons with initial energy 50 eV (panel A), 100 eV (panel B), 250 eV (panel C), 500 eV (panel D), 1,000 eV (panel E) and 5,000 eV (panel F). Dashed line: simulations using the ORNL version of the dielectric function and the Born approximation; solid line: simulations using the ECN model of the dielectric function and the Born approximation; dotted line: simulations using the ECN model of the dielectric function with Born corrections (see text).

electrons in liquid water have been presented based on the ECN model of the dielectric function, which is used to calculate inelastic cross section beyond the Born approximation. Theoretical developments in inelastic scattering are discussed and their effect on track structure is investigated. The results reveal that the optical data and dispersion relationships used in semi-empirical models of the dielectric function, as well as the corrections to the Born approximation, may have a sizeable effect on track structure at the nanometer scale (DNA level) where the details of low-energy transport are most critical. It was also

found that, in some cases, the influence of non-RPA and non-Born effects cancels out because of their opposite effect on the inelastic mean free path. Although the approach based on the dielectric function and the Born approximation is well established for calculating inelastic cross sections for amorphous media (such as liquid water), its application to MCTS simulations requires careful implementation and further approximations to properly account for secondary electron production and transport down to the threshold energy of the individual energy-loss channels.

APPENDIX A

The ordinary and derivative Drude functions are defined, respectively, by

$$D(E; E_j) = \frac{f_j E \gamma_j}{(E_j^2 - E^2)^2 + (E \gamma_j)^2}, \quad (A1)$$

$$\tilde{D}(E; E_j) = \frac{2f_j (E \gamma_j)^3}{[(E_j^2 - E^2)^2 + (E \gamma_j)^2]^2}, \quad (A2)$$

and their Kramers-Kronig (K-K) pairs are

$$D_{K-K}(E; E_j) = \frac{f_j (E_j^2 - E^2)}{(E_j^2 - E^2)^2 + (E \gamma_j)^2}, \quad (A3)$$

$$\tilde{D}_{K-K}(E; E_k) = \frac{f_k (E_k^2 - E^2) [(E_k^2 - E^2)^2 + 3E^2 \gamma_k^2]}{[(E_k^2 - E^2)^2 + (E \gamma_k)^2]^2}. \quad (A4)$$

APPENDIX B

The contributions $F_{kl}(E)$ are calculated by the sum of two terms related to the truncated and smoothing parts of higher level Drude functions. So, for an ionization shell j we write:

$$F_j(E) = F_j^l(E) + F_j^ll(E), \quad (B1)$$

where $F_j^l(E)$ are contributions from the truncated part of higher energy Drude functions

$$F_j^l(E) = \sum_{n=j+1}^{j_{\max}} [D(E; E_n) \Theta(B_n - E) w_j^l(E)], \quad (B2)$$

with the weighting factor $w_j^l(E)$ determined by

$$w_j^l(E) = \frac{D(E; E_j)}{\sum_{i=1}^j D(E; E_i) H(E - B_i)}, \quad (B3)$$

and $F_j^ll(E)$ are contributions from the smoothing part of higher energy Drude functions

$$F_j^ll(E) = \sum_{n=j+1}^{j_{\max}} D(E; B_n) \exp(B_n - E) H(E - B_n) w_j^ll(E) \quad (B4)$$

with the weighting factor $w_j^ll(E)$ determined by

$$w_j^ll(E) = \frac{D(E; E_j)}{\sum_{i=1}^j D(E; E_i)}. \quad (B5)$$

Note that the two weighting factors, $w_j^l(E)$ and $w_j^ll(E)$ are similar but not identical since the truncated part of a particular Drude function can only be distributed to ionization shells of lower (binding) energy. Similarly, for an excitation level k we write:

$$F_k(E) = F_k^l(E) + F_k^ll(E), \quad (B6)$$

where $F_k^l(E)$ denotes the contribution from the truncated part of the Drude functions that falls below the first ionization potential

$$F_k^l(E) = \sum_{j=1}^{j_{\max}} D(E; E_j) \Theta(B_{j=1} - E) w_k^l(E), \quad (B7)$$

with the weighting factor $w_k^l(E)$ determined by

$$w_k^l(E) = \frac{\tilde{D}(E; E_k) \Theta(B_{j=1} - E_k)}{\sum_{i=1}^{k_{\max}} \tilde{D}(E; E_i) \Theta(B_{j=1} - E_i)}, \quad (B8)$$

and $F_k^ll(E)$ is the contribution from the smoothing part of the Drude function of the first ionization shell

$$F_k^ll(E) = D(B_{j=1}; E_{j=1}) \exp(B_{j=1} - E) H(E - B_{j=1}) w_k^ll(E), \quad (B9)$$

with the weighting factor $w_k^ll(E)$ determined by

$$w_k^ll(E) = \frac{\tilde{D}(E; E_k)}{\sum_{i=1}^{k_{\max}} \tilde{D}(E; E_i)}. \quad (B10)$$

In the above expressions, $\Theta(\dots)$ and $H(\dots)$ are unit-step functions with $\Theta(0) = 1$ and $H(0) = 0$.

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